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Knowledge Graphs for Representing Knowledge Progression of Students across Heterogeneous Learning Systems

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Abstract

Student models are inevitable in any educational system for analysis, reasoning, and making decisions. Courses and lessons from multiple heterogeneous learning systems may be present in the entire learning trajectory of students. So, the knowledge states of the students and domain knowledge from various learning systems need to be collated to guide the students through their learning path. Addressing this issue, we propose a 3-layer *Student Knowledge Graph* as an advancement of the educational Domain Knowledge Graphs to represent the knowledge progression of students. The first two layers of the Student KG are a semantic network of the performance history of students, and *Knowledge Components* from multiple learning systems. A knowledge layer of public knowledge base and domain ontology was added to interconnect the *Knowledge Components* of various learning systems by linking them to the entities of the knowledge layer. The results show that a clear and precise description of the *Knowledge Components* produces better connectivity between learning systems and the knowledge layer through text-based entity linking. To demonstrate the applicability of *Student Knowledge Graph*, the knowledge gaps of the Computer Science students of a University system in India were analyzed, and the remedial recommendations were generated from self-paced learning system datasets. We submit that including the knowledge states of the students in educational knowledge graphs provides an effective ranking and, thereby, personalized recommendations when compared to Educational Domain Knowledge Graphs.

Keywords Heterogeneous learning systems · Student knowledge progression · Knowledge graphs · Knowledge gap analysis · Remedial recommendations

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Introduction

Student models are an inevitable component of any educational system, especially in the current era of technology-enhanced learning environments. The current student models are more focused on supporting the systems to make decisions within an academic course, program, or learning system than assisting the students to keep track of their entire learning progression. So, the students remain unaware of their possibilities or knowledge gaps in the trajectory for achieving a specific goal. To guide the students in their learning path, the knowledge progression of students needs to be integrated and linked with external knowledge from various systems. The communication between the student models and domain knowledge models is difficult due to the heterogeneity of learning systems (Baker, 2019). Collating and sharing student knowledge across heterogeneous learning systems requires a semantic representation of the same knowledge in a global context.

A Knowledge Graph (KG) is a semantic network that illustrates the relationship between real-world entities. Resource Description Framework (RDF) vocabularies and Web Ontology Language (OWL) formalize the semantic schema of KG, enabling the reasoner applications to derive new knowledge from it Hogan et al. (2021). The KG was exploited as a tool for tracing the knowledge state of students in tutoring systems which shows the capabilities of KG for analyzing student performance (Nakagawa et al., 2019). In recent studies, the knowledge of the educational domains was represented as a KG of learning objects and their relations for guiding students through their learning paths (Shi et al., 2020). However, relating the domain knowledge from two educational systems requires an agreement between them due to the heterogeneous and evolving nature of knowledge. The domain knowledge of learning systems can be represented by a semantic network of Knowledge Components (KC). KCs are fragments of persistent, domain-specific information like concepts, principles, or facts that should be used to accomplish educational tasks (VanLehn, 2006). Educational Domain KGs were created by identifying KCs and the relations between them (Chen et al., 2018). Identifying the relationship between KCs from various learning systems may end up in repetition and ambiguity. So, linking the KCs of learning systems to existing knowledge bases or domain ontologies might help remove such ambiguities. Domain ontologies are special kinds of knowledge bases that represent the abstract model of a domain including the domain concepts, their relations, and restrictions (Stancin et al., 2020). Knowledge bases could be connected to the domain knowledge of learning systems exploiting entity linking. Entity linking is an approach that maps the relevant mentions in the raw text to the entities in the knowledge bases (Oliveira et al., 2021; Shen et al., 2014). However, connecting the learning systems is not enough, the need/benefits of such linking should be demonstrated. So, considering the mentioned possibilities of KG and related techniques, this paper addresses the following research questions:

1. RQ 1: How effective is entity linking to connect the domain knowledge (KCs) across heterogeneous learning systems and represent the student knowledge progression?

2. RQ 2: How effective is the Knowledge Graph (KG) in representing the knowledge gaps of the students?
3. RQ 3: Can Knowledge Graph (KG) find the self-paced courses related to the knowledge gaps of the students across heterogeneous learning systems?

To address the above-mentioned research questions we designed a formal ontology named *Student Knowledge Representation ontology* that can represent a 3-layer semantic network called Student KG. Student KG contains three layers where student information (evidence for student performance) in the top layer, corresponding KCs of learning systems in the middle layer, and the knowledge base in the bottom layer which acts as a mediator for all learning systems in the middle layer. The middle layer holds the domain knowledge of the learning systems supported by the knowledge base in the third layer and both of these layers form Domain KG. Our results show that the Domain KGs of multiple learning systems can be connected meaningfully by linking KCs to the knowledge base. We show that Student KG is effective in representing the progression of student knowledge and analyzing the knowledge gaps. We demonstrate the applications of Student KG through the use-case of remedial course recommendations that include multiple learning systems. Student KG provided better diversity and relevance-based ranking in course recommendations for students than Domain KG.

Contributions

1. A knowledge architecture to relate the domain knowledge of learning systems employing knowledge bases without affecting or altering the functioning of their student and domain models
2. Knowledge Graphs for representing student knowledge progression and knowledge gap analysis across heterogeneous learning systems
3. Student Knowledge Graphs for personalized Remedial Course recommendations

Paper Organization

In the next section, we provide an overview of the existing literature on student knowledge representation models, knowledge graphs, knowledge gap analysis, and remedial recommender systems. The third section provides a detailed description of our Student KG model in representing student knowledge and applying it in knowledge gap analysis, and remedial course identification. The fourth section exhibits the results of our knowledge representation methods followed by a discussion on the results, limitations, and future possibilities of the approach. The conclusion section summarizes our research and contributions to the research literature.

Literature Review

The literature presents a great deal of effort in representing and modeling student knowledge. A significant amount of work has been done in making recommendations to the students based on their past performances and knowledge gaps. In attempting

to link heterogeneous learning systems, we explored the various knowledge models and drew inspiration from techniques proposed for linking entities in a range of fields and domains. We discuss in this section, the current literature which inspired us to propose and work on the approach presented in this paper.

A web-based unique identifier for each KC is needed to represent the domain knowledge globally. The World Wide Web Consortium has provided various standards and vocabularies for representing knowledge semantically which include Resource Description Framework (RDF), Simple Knowledge Organization System (SKOS), SPARQL query language, and Web Ontology Language (OWL). RDF is a model for exchanging data on the web by representing data as entities and their relationships (Brickley & Guha, 2014). SKOS is a namespace, built on the top of RDF that defines a vocabulary for sharing and linking knowledge across the web using semantics (Miles & Bechhofer, 2009). Dublin Core Metadata Initiative (DCMI) is another namespace for creating profiles of resources in semantic web (Board, 2020). OWL is built on top of RDF and used to define ontologies for things, groups, and their relations (Boris et al., 2012). RDF, OWL, SKOS, and DCMI are the standards and taxonomies that define the schema for various types of web resources (Kusuma et al., 2022). The standards and taxonomies mentioned above contain more generic terms like a resource, concept, subject, etc., and the learning systems require more specific terms including Course, student, performance, learning system, and KC. So, a taxonomy ontology that includes and inherits the existing standards and taxonomies is required to represent the knowledge in learning systems.

Representation of Domain Knowledge and Student Knowledge Progression in Learning Systems

Ontology is a popular knowledge representation method in the educational domain because it can formulate the instructional concepts, their properties, and relationships in a domain semantically (Stancin et al., 2020). Many research works show that ontology is an effective tool for representing the skills and learning styles of students to recommend learning paths or learning resources (Obeid et al., 2018; Tarus et al., 2017; George & Lal, 2019; Clemente et al., 2022). Most of the recent research on ontologies for educational domain modeling or student modeling focuses on recommender systems. However, the recommendations are from within a system or similar systems.

EduProgression is an ontology-based semantic representation designed for the knowledge progression of the French elementary school curriculum (Rodríguez Rocha et al., 2017). EduProgression formalizes the progression of academic programs as three Cycles ranging from learning fundamentals to deepening each Element of Knowledge and Skill. KnowEdu is a system for creating educational KGs by identifying concepts and relations from educational text (Shi et al., 2020).

ON-SMMILE is a student model based on an ontology network that describes the student profile, knowledge states, knowledge objects, and traces of student activities (Yago et al., 2018). EDUC8 is another ontology that models the learning trajectory of students and their economic events to achieve data interoperability between various higher educational institutions (Iatrellis et al., 2019). OntoSIDES is an ontology

designed for the student progress monitoring KG for the national evaluation system of French medical schools (Palombi et al., 2019). The outcome of questions for students was predicted using OntoSIDES KG (Ettorre et al., 2020). However, these models focus only on the same system or systems that follow the same protocols, and terminology. Nowadays students learn from various offline and online systems. Our Research question RQ 1 focuses on maintaining the continuity of knowledge progression of students across learning systems. So, inspired by the above-mentioned models, we contribute to the existing literature by creating a KG for the knowledge representation of students in various learning systems by interlinking their KCs using the knowledge layer.

Educational KGs were generally created to represent the knowledge of one or more educational domains in terms of educational concepts, resources, and their relations. The educational KGs are knowledge bases created from various forms of information like text, images, and structured data and formalized by ontology. MOOC-KG is recent research on the domain knowledge of MOOC courses for MOOC (Massive Open Online Courses) learning resource utilization (Dang et al., 2019). CKGG is a KG created for Chinese high school geography educational resource representation (Shen et al., 2021). Linking the mentions of KCs in educational text with their corresponding entities in a knowledge base/KG is an Entity Linking task (Shen et al., 2014). EDUKG is a KG generated from heterogeneous data for Chinese K-12 educational knowledge representation with a text-based entity linking mechanism to relate the educational texts to the entities in EDUKG (Zhao et al., 2022). These KGs are created to represent either MOOC courses or traditional educational courses. These research articles do not discuss sharing the knowledge progression of the students or relating the domain knowledge across heterogeneous learning systems.

Entity Linking in Educational Knowledge Graph

According to RQ1, the effectiveness of *the entity-linking* approach has been experimented with in our research. Many text-based and graph-based methods are used for entity linking. Most text-based linking methods are based on semantic matching using word embeddings (Nguyen et al., 2019; Fernandez et al., 2018; Deepak et al., 2020). Text-based semantic matching was improved using structural information based on graphs (Xie et al., 2020). Graph-based methods are built on the extraction of subgraphs with relevant candidates, ranking the entities using graph embeddings, and computing entity relatedness within the graph (Cifariello et al., 2019; Parravicini et al., 2019; Ponza et al., 2020). LiTeWi combines unsupervised term extraction and entity linking algorithms for generating educational ontologies (Conde et al., 2016). EduProgression links each *Element of Knowledge and Skill* with the related thesaurus for identifying learning resources related to the *Element of Knowledge and Skill* using the knowledge base named DBPedia (Lehmann et al., 2015; Rodríguez Rocha et al., 2017). DBpedia spotlight is developed for entity disambiguation by identifying the mentions of DBpedia entities in the given input text (Mendes et al., 2011). We have chosen DBPedia as the third layer component as it was already exploited to represent educational knowledge progression. However, a knowledge base contains many more entities and

relations than educational concepts and their relations. Narrowing down the knowledge base entities and relations to valid educational concepts and relations causes computational overhead. So, we compared entity linking with DBpedia and Computer Science Ontology (CSO), which is generated from academic articles for representing the concepts and their relations in the computer science domain (Salatino et al., 2018). The Computer Science Ontology (CSO) is a large-scale ontology of academic topics that provides an unsupervised text classification model created for classifying an article's title, abstract, and keywords into a set of computer science topics with a URI (Salatino et al., 2019, 2018). We have chosen CSO for our experiments as the third layer (an alternative for DBpedia) because our research is based on computer science courses and CSO has its classification mechanism to identify topics from raw text. In addition, CSO contains educational relations like 'SuperTopic of' which helps identify the area of study in which the student is weak. It is difficult to find a domain ontology that occupies every topic from the computer science domain. Our research points out both the need and benefits of ontologies specific to educational domains.

Analysis of Knowledge Gaps of Students and Recommendations

While creating student knowledge progression as a shareable KG it is vital to discuss whether the shared student model can reason and intervene beneficially among the learner community (Baker et al., 2021). A benefit of the shareable student model is personalizing the decisions across various learning systems that help the learners achieve the lifelong learning goals (Kay, 2008). So, we consider the knowledge gap analysis of the individuals and personalized remedial recommendations from the course list of another system, as the use cases of Student KG.

In intelligent tutoring systems, the knowledge gaps are identified by verifying the mastery level of the students against each KC. The students mastered a particular skill if their mastery level is 'Mastered'. Knowledge tracing models were created for verifying the mastery of students on a concept/skill (Haridas et al., 2019; Gutjahr et al., 2021). Recently a recommendation system was proposed to recommend the prerequisite resources for the KC to the students if they did not master the KC (Sablayrolles et al., 2022). Nacelle is a skill-gap analysis model for the workplace enabled by the conversational KG which analyzes the knowledge gaps by comparing the answers given by the participant with the expected answers (Loh et al., 2022). Graph algorithms like PageRank, Node Similarity, and Community detection were employed for skill-gap analysis of KG that includes the skill profile of engineering employees (Choi et al., 2021). KG and graph algorithms were also instrumented for skill analysis in the labor market (Deepak et al., 2020). The knowledge gap analysis research papers mentioned in this section analyze the low performance in specific skills and do not inform about the broader areas in which the student has difficulty. Another research proposed a content-based recommendation system for learners from various MOOC (Massive Open Online Courses) systems by identifying topics similar to the contents learned by the students using Non-negative matrix Factorization (Campos et al., 2022). Some of the recent models focus on the competency of students to recommend learning resources and learning pathways (Diwan et al., 2021; Sablayrolles et al., 2022;

Songer et al., 2020). Navigated learning is a recent approach for guiding the learning process that locates and curates student competency, mediates learning pathways, and facilitates the learning resources for the students (Songer et al., 2020). However, the implementation and execution of Navigated Learning was manual without using technology. Following the Navigated Learning Concept, a 2-dimensional navigable progression space (competency map) was created, using instructional topics as the X-axis and the complexity of the topic as the Y-axis (Diwan et al., 2021). This competency map is a collection of learning pathways to achieve certain competencies which present both the distance between competencies and the partial order of the competency sets. Knowledge Maps and KG were employed to recommend the learning paths to the students to learn a particular KC (Shi et al., 2020; Zhu et al., 2018). Instead of automatically assigning the KC that the students want to learn, the KC was provided as input to both models. Another recent recommendation System based on the competency of students was proposed exploiting ON-SMMILE ontology (Clemente et al., 2022). This recommendation system suggests learning activities to the students in an adaptive learning environment by recognizing the positive and negative qualities of their previous activities. The studies mentioned in this subsection do not consider the learning history of the student from the previous learning system which might help to avoid the cold start problem in the recommended systems. Recent work has shown that linked open data could resolve the cold start problem in recommender systems (Natarajan et al., 2020). So, it is crucial to ensure that the new learning system can identify the knowledge gaps of students based on their learning history from the previous learning system and provide remedial recommendations.

We follow the ontology-based approach to represent the student knowledge progression. Unlike the above-mentioned models, we expand the KG across the courses from formal learning and self-paced learning courses. We attempt to identify the knowledge gaps from the KG of the performance history of students that cover multiple courses. Our model performs the knowledge gap analysis and automatically extracts the remedial course list from the KG. We also analyze the difference in recommendations when the students' performance history is not included in the KG.

Methodology

Multiple factors need to be considered while representing the knowledge profile of students globally. Students learn from multiple educational systems ranging from traditional classrooms to self-paced courses. So integrating the KCs that represent the domain knowledge of different learning systems is most critical. The KCs across various learning systems which differ in text, may hold the same meaning or contribute to each other. So, the relationship between the KCs across the learning systems should be recognized and maintained in the *knowledge profile* of students. Besides the KCs, the *performance evidence* of students is required to depict their knowledge level in each KC. The *previous scores* of students in a topic play an important role in predicting future results (Haridas et al., 2020). Another important information is the complexity

of the questions which shows the *cognitive level* of the student in a specific KC. The *time of evaluation* of the student performance is also vital for maintaining the order of knowledge acquisition and capturing recent advancements in the learning progression. We attempt to construct Student KG as a 3-layer semantic network of the above-mentioned components of student knowledge to represent them in a global context.

The KG was created for Computer Science students and formalized by SKR (Student Knowledge Representation) ontology. Figure 1 shows the 3-layer organization of the components in Student KG. The performance layer and the learning system layer are represented using SKR ontology, and domain ontologies and knowledge bases represent the knowledge layer. In the performance layer, the students have multiple performance entities from various learning systems and the performance could be high, medium, or low depending on the score earned by the students. The complexity of the questions is a property of the performance entity. The performance evaluation is an event that captures the knowledge progression of students which is represented as the Date property in the performance node. The learning system layer consists of the KCs from multiple learning systems. Each performance is evaluated for one of the KCs in the middle layer that represents domain knowledge of various learning systems. The relationship between the KCs within a learning system may or may not be defined. So, Fig. 1 shows the KCs within the learning system as disconnected entities. The bottom layer is the knowledge layer that contains entities in the public knowledge bases that define the relations between various computer science topics.

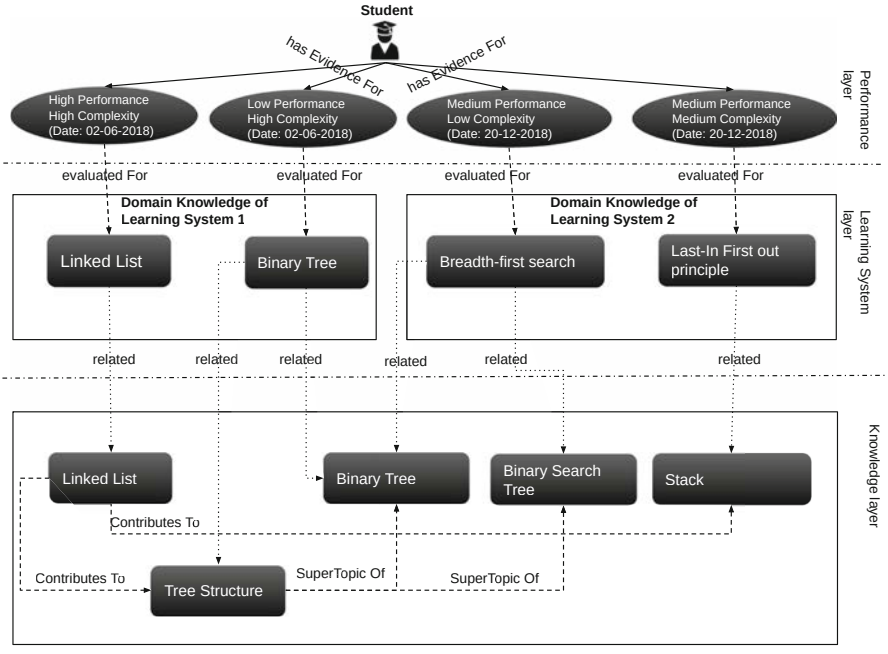


Fig. 1 The knowledge organization in student KG

In this section, we describe the datasets utilized in the experiment, the implementation of Student KG using SKR ontology, and the use cases of Student KG in performance analysis and course recommendations for students.

Datasets

The ways of maintaining the KCs in various courses or educational systems are different. So, We considered a dataset of courses from formal education that contains a variety of Knowledge Components. Some of the courses provide the KCs as fine-grained concepts whereas some of the courses provide higher-level learning objectives which may not contain all concepts in the text. In such cases additional text is required, So, we included the content of questions corresponding to the learning objective. The examples are given in Table 1.

Our experiment is the knowledge representation across heterogeneous systems. So, we considered the datasets from traditional and online systems to address hetero-

Table 1 The samples of knowledge components

Course	KC	Questions
Applications of machine learning	Clustering and its applications. Learn various types of clustering	Suppose a cluster consists of points (5, 2), (6, -1), and (7, 0). How does the BFR algorithm represent this cluster? State the benefits of such a representation
		Show at least four iterations of hierarchical clustering to produce dendrograms. Two clusters are merged using Inter-cluster distance = minimum of the distances between any two points, one from each cluster
		Prove that the time complexity of the K-means clustering algorithm is less than Hierarchical clustering. Prove that the time complexity of the K-means clustering algorithm is less than Hierarchical clustering
		How does the CURE clustering algorithm overcome the drawback of conventional hierarchical clustering?
		Explain the significance of Mahalanobis distance in BFR clustering
		What do you mean by subspace clustering? Explain the need for such a clustering

geneity in the types of learning systems. The publicly available traditional education datasets do not have the data of continuous evaluation results of two semesters for the same students along with the KCs. So, we utilized the student performance data of undergraduate and graduate students of 70 computer science courses from the 2018-2019 academic year, AMRITA University, Amritapuri Campus, India. The dataset contains two semesters' performance data of 1381 students enrolled in more than one course from 2 continuous semesters. The student details in the datasets are completely anonymized. We did not collect any personal data of students other than performance data. The results of 32 courses of the first semester were considered for Knowledge Gap Analysis. We generated the student KG of 1381 students from the 32 courses of the first semester. The other 38 courses of the second semester were utilized to verify the impact of the knowledge gaps of students in the course performances of the second semester. The samples of records in the university student dataset are given in Table 2. The first example in the Table 2 should be understood in the following way. Student '1' obtained a CCSS score of 1.0 (perfect score) for all questions with the cognitive complexity of 'Analysis' which belongs to the KC 'Language Models' in Natural Language Processing course.

In addition to the university dataset, two self-paced course datasets were instrumented to show the link between multiple learning systems: The Coursera course dataset and the Udacity course dataset. The Udacity dataset contains a collection of 262 self-paced courses and the skills taught in the courses (Ayush, 2021). The Kaggle Coursera course dataset has 3416 courses and skills associated with each course (Khushee, 2022).

RQ1:Linking KCs from Various Learning Systems with the Knowledge Layer

The connection between KCs from various learning systems and the knowledge layer was achieved by text-based entity linking. We have considered the candidate entities from CSO ontology and DBpedia for linking the KCs in various learning systems. Both of the knowledge bases have their entity-linking mechanisms. We chose CSO for our research as it contains Computer Science Concepts and their relations only. However, We included DBpedia as an alternative because CSO does not contain all topics from computer science. The description of each KC and the corresponding exam questions were considered as the input text for entity linking. CSO unsupervised classifier was

Table 2 A sample university student examination dataset for creating student KG (The values for Student ID and Exam Date are placeholders)

Student Id	Course	KC	Cognitive complexity	Exam date	CCSS
1	Natural language processing	Language models	Analysis	12-12-2019	1.0
2	Natural language processing	Disambiguate sentences	Creation	12-12-2019	0.25

instrumented for identifying semantic equivalent topics for the input text from CSO ontology whereas the semantic equivalents from DBpedia were identified by exploiting DBpedia spotlight.

The *relation* and *subject* properties in DCMI terms were employed to represent the relations between a KC and the related entity. *Relation* is a property that connects the two related resources. The subject property defines the relation between the resource and its subject. In DBpedia, the entities are represented as resources and the resources are mapped with the concepts (instances of SKOS Concept) using the *subject* property. So the relation between KC and DBpedia resources was mapped using *the relation* property as shown in Fig. 2.

CSO was developed as an extension of SKOS ontology where the *Topic* class in CSO is a subclass of the Concept class in SKOS, enabling linking with the instances of any subclass of Concept. The *Course* and *KnowledgeComponent* classes in SKR are also created as the subclasses of the SKOS Concept class. So all relations between two Concept instances defined in SKOS can exist between the CSO Topic instances and the SKR Course and KnowledgeComponent instances. We used the DCMI term 'subject' to represent the relationship between SKR *KnowledgeComponent* and CSO *Topic* as shown in Fig. 2.

Ontology for Student KG

Student KG is a collection of triples <H, R, T> (Head, Relation, Tail) where each triple <H, R, T > denotes the directed relation r from entity h to entity t. The classes and properties of *Student Knowledge Representation* ontology that represent <H, R, T> are shown in the Fig. 3. We have considered the minimum set of node types and

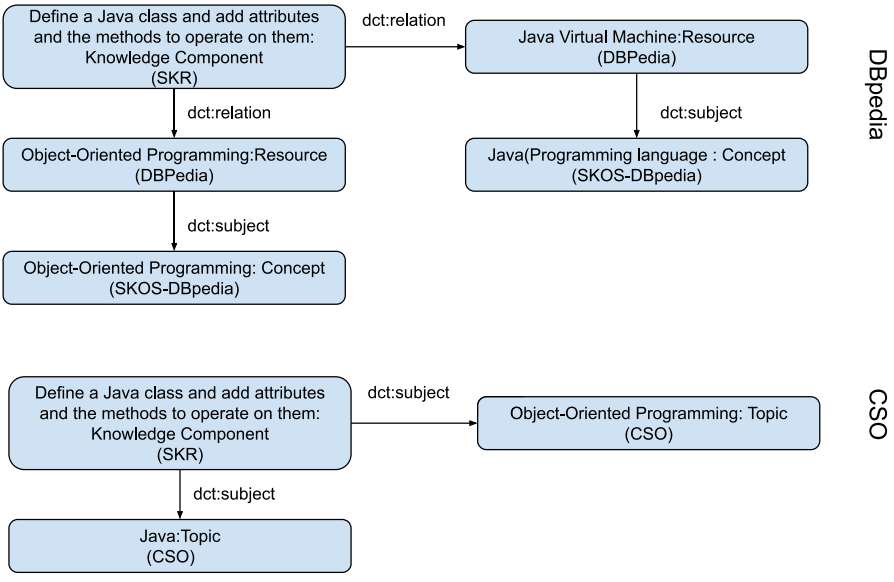


Fig. 2 The entity linking examples with DBpedia and CSO

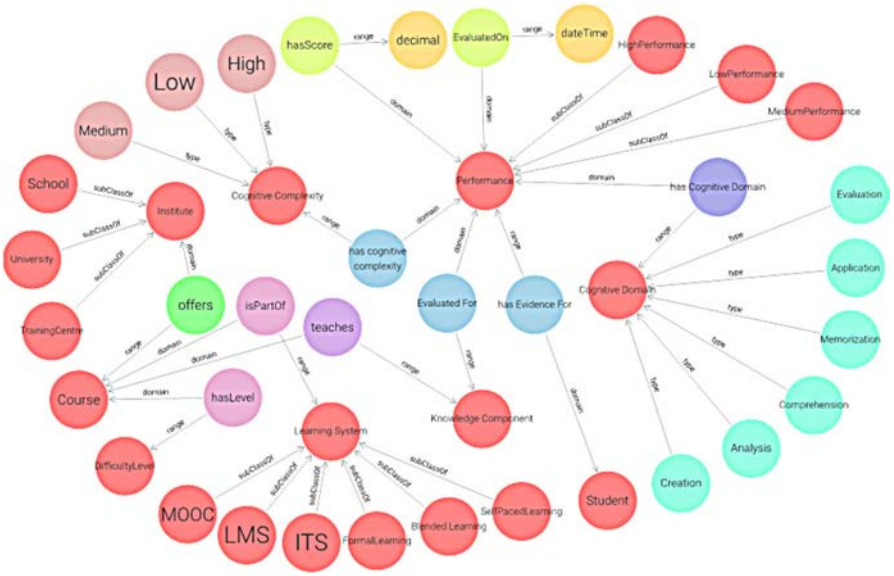


Fig. 3 SKR ontology

properties for representing student knowledge. In Fig. 3, Each 'Relation' has at least a domain node and Range node. For example, the Relation *hasEvidenceFor* is the relation between the Student node (Head), and the Performance Node (Tail). The Performance node can either be the *HighPerformance*, *MediumPerformance*, or *LowPerformance* node which are the sub-classes of the Performance node. Each student has evidence for their performance for a KC or Course. This relation is represented using the *Evaluated For* relation between the *Performance* node and *KnowledgeComponent* node. Another example is the *teaches* relation whose domain is *Course* and range is *KnowledgeComponent* nodes. Instances of each node are connected to the node with a 'type' relation.

Each performance node has a Context-bound Cognitive Skill Score represented by *hasSCore* attribute, (CCSS) (MD & Krishnamoorthy, 2022) computed for each KC, and attributed to the Cognitive Complexity of questions attempted by the student. The CCSS scores for students were calculated by grouping the examination questions based on cognitive complexity and averaging the scores earned by the students. The computation of CCSS scores is exactly as mentioned in the referenced paper (MD & Krishnamoorthy, 2022). The performance is high if the score is greater than 0.7, medium if the score is between 0.4 and 0.7, and low if the score is less than 0.4. The Cognitive Complexity of the Performance is Low if the Cognitive Domain of the associated questions is Memorization or Comprehension, Medium if the Cognitive Domain is Application or Analysis, and High if the Cognitive Domain is Evaluation or Creation. An institute (School, University, or Training Center) offers a Course that is part of a Learning System and teaches a set of KCs. *FormalLearning*, *SelfPacedLearning*, and *BlendedLearning* are the subclasses of the class *LearningSystem*.The algorithm for creating some important nodes and relations of the Student KG is given in Algorithms 1, 2, and 3.

Algorithm 1 Algorithm for creating student KG.

```
Inputs————————begin-
Rows = The student performance data in the format of Table 2

URI=URL of the SKR ontology
StudentKG=[]
SKR=Student Knowledge Representation Ontology
LS=LearningSystemID
LSType=LearningSystemType
Inputs————————end-
Output : Student KG
for each row R in Rows do
    Timestamp =get UTC timestamp
    StudentID = R[StudentId]
    Course = R[Course]
    KC = R[KC]
    CC =R[Cognitive Complexity]
    ED = R[Exam Date]
    CCSS =R[CCSS]
    nodeID=URI+StudentID
    S=createNode (nodeID,studentKG,SKR['Student'])
    nodeID=URI+Course
    CSN=createNode (nodeID,studentKG, SKR['Course'])
    nodeID=URI+KC
    KCN=createNode (nodeID,studentKG,SKR['Knowledge Component'])
    nodeID=URI+CC
    CCN=createNode (nodeID,studentKG, SKR['Cognitive Complexity'])
    nodeID=URI+LS
    LSN=createNode (nodeID,studentKG,SKR['LSType'])
    PerType='LowPerformance'
    if ROWS[CCSS]>= 0.7 then
        PerType='HighPerformance'
    end if
    if ROWS[CCSS]<0.7 and OWS[CCSS]>0.4 then
        PerType='MediumPerformance'
    end if
    nodeId=URI+PerType+Timestamp
    performN=createNode (nodeID,studentKG,SKR[perType])
    performN[EvaluatedOn]=R[Exam Date]
    performN[Score]=CCSS
    addEdge (S,URI+'hasEvidenceFor',performN)
    addEdge (performN,URI+EvaluatedFor,KCN)
    addEdge (Course,URI+'teaches',KCN)
    addEdge (performN,URI+'hasCognitiveComplexity',CCN)
    addEdge (LSN,URI+'Offers',Course)
end for
return StudentKG
```

The Student KG is a collection of generated triples that are in the form <Head, Relation, Tail> (<H, R, T>). Table 3 gives some of the generated triples. Each Head, Relation, and Tail has a Unique Resource Identifier (URI). The Head and Tail have the URIs to two nodes and the Relation (Predicate) is

Algorithm 2 Algorithm for creating a node.

```
INPUT=nodeID,studentKG, NodeType
OUTPUT=node
Function: createNode
Set node to Null
if nodeId not in StudentKG  then
    node=new Node (nodeID)
    node[Type]=NodeType
    ADD node to StudentKG
else
    node=Student KG[nodeId]
end if
return node
```

Algorithm 3 Algorithm to ADD an edge to student KG.

```
INPUT: Node1, Node2, Relation,StudentKG
OUTPUT: StudentKG
Function:addEdge
Triple = (Node1,Relation,Node2)
if Triple does not exist in Student KG then
    ADD Triple to StudentKG
end if
return StudentKG
```

the URI that represents the relation between the Head and Tail nodes. For example, the HTTP URI that represents SKR# 8a8383456788d6e001679632ff1114e4 is http://<URL-TO-SKR>/ 8a8383456788d6e001679632ff1114e4, and the URI for the predicate RDFS#type is http://<URL-TO-RDFS>/type. The attributes of the nodes are represented as <H, R, t> where t is a numeric, string, or date value. SKR#8a8383456788d6e001679632ff1114e4 is a Unique ID generated for the student to anonymize the name. Similarly, there are unique IDs generated for the nodes of KnowledgeComponent, Course, University, etc.

Querying Student KG for Knowledge Gap Analysis

We have created the KG by creating the triples using algorithms mentioned in the previous section and uploaded the triples to GraphDB (Ontotext, 2024). The knowledge gap analysis was implemented using SPARQL queries against the 3-layer Student KG. The low-performance subgraphs for all students were extracted using SPARQL queries followed by computing the in-degree of each concept. The in-degree of a node in a directed graph is the number of incoming edges (relations) of a node. According to Student KG, the low-performance subgraph of a student has only the low-performance nodes that have edges to KCs and from KCs to topics. So the number of edges to the topics shows the number of times the student scored low on a topic. The low-performance subgraph topics were extracted using the SPARQL query equivalent to the Algorithm

Table 3 The sample of triples in student KG

Head (Subject)	Relation (Predicate)	Tail (Object)
SKR# 8a8383456788d6e001679632ff1114e4	RDF#type	SKR#Student
SKR#0513752c-2a84-4ffa-842d-8b435f7f2a15	RDF#type	SKR#KnowledgeComponent
SKR#00fa6e17-69a8-4e58-9a87-6c2ef5992869	RDF#type	SKR#Course
SKR#001_university	RDF#type	SKR#University
SKR# 42880aea-5511-4abb-9c90-921e55295c80	RDF#type	SKR#HighPerformance
SKR#001_university	RDF#type	SKR#University
SKR#0513752c-2a84-4ffa-842d-8b435f7f2a15	RDFS# label	Model evaluation
SKR# 00fa6e17-69a8-4e58-9a87-6c2ef5992869	RDFS# label	Application of machine learning
SKR# 8a8383456788d6e001679632ff1114e4	SKR#hasEvidenceFor	SKR#42880aea-5511-4abb-9c90-921e55295c80
SKR# 42880aea-5511-4abb-9c90-921e55295c80	SKR#hasScore	8.0
SKR# 42880aea-5511-4abb-9c90-921e55295c80	SKR#EvaluatedOn	12-02-2018
SKR# 42880aea-5511-4abb-9c90-921e55295c80	SKR#EvaluatedFor	SKR#0513752c-2a84-4ffa-842d-8b435f7f2a15
SKR#00fa6e17-69a8-4e58-9a87-6c2ef5992869	SKR#teaches	SKR#0513752c-2a84-4ffa-842d-8b435f7f2a15
SKR#001_university	SKR#offers	SKR# 00fa6e17-69a8-4e58-9a87-6c2ef5992869
SKR# 00fa6e17-69a8-4e58-9a87-6c2ef5992869	DCMI#subject	https://cso.kmi.open.ac.uk/topics/machine_learning
SKR#0513752c-2a84-4ffa-842d-8b435f7f2a15	DCMI#subject	https://cso.kmi.open.ac.uk/topics/association_rules

4. In Algorithm 4 the limit variable can be set to any integer. For the experiment, we set it to 3 and 5 because we are using a semester's data for creating the KG, and thus the chances of having more links between a low-performance node to a concept is less. The number of occurrences of each concept was computed and sorted to identify the student's weak areas.

Identifying Courses Related to the Knowledge Gap of Students

The topics in which the students are weak were identified from the low-performance subgraph of students. The self-paced courses that are linked to the weak areas of students were extracted for recommendation to the students. The remedial list of courses was extracted as specified in Algorithm 5. Get the self-paced courses and

Algorithm 4 Algorithm for knowledge gap identification.

```
Function: GetLowPerformanceConcepts
Input: StudentID, limit
Output: All Concepts where the student has a Knowledge Gap
LPNodes=Get all Nodes with type SKR#LowPerformance where StudentID node has a relation SKR#hasEvidenceFor.
LPKClist=Get All KCs where LPNode has a directed connection.
LPConcepts=Count all Concepts that have a path from each KC in LPKClist Group by each Concept where count >= limit.
return LPConcepts
```

count (wt) of distinct weak topics that are linked to the self-paced courses from KG. The list of courses was prioritized based on the descending order of the count (wt).

Algorithm 5 Algorithm for identification of courses related to the knowledge gaps of the student.

```
Function: GetCourses
Input: StudentID
Concepts= GetLowPerformanceConcepts (StudentID)
Set SelectedCourses to empty list
Courses=Get All H of type Cousre which has a triple <H,SKR#partOf,LearningSystem> where LearningSystem of type SKR#SelfPacedLearning
for each concept in Concepts do
    selCourses= Get all H which has a triple < H, DCMI# subject, concept> in Courses
    Add selCourses to SelectedCourses
end for
return SelectedCourses
```

Evaluation

We have evaluated each Research Question Separately. The evaluation methods are described in the following subsections.

Evaluation Of RQ1

The effectiveness of entity linking for relating the knowledge components of the courses to the knowledge layer is identified by the precision and recall of the linked topics from DBpedia and CSO. Precision and recall for DBpedia are given below.
Precision=Correct linked topics from DBpedia/Total expected topics
Recall=Correct linked topics from DBpedia/ all linked topics from DBpedia
The total expected topics were manually identified from the questions and precision computes how many of them correctly identified. Recall shows how many topics were correct from the total links obtained by entity linking. The incorrectly linked topics are the entities identified by the entity linking approach and do not belong to the Total expected topics.

We chose a machine learning course because the majority of the machine learning topics are present in the knowledge bases. We manually identified the expected topics from the input text related to each KC and verified how many of them have been identified by DBpedia spotlight and CSO classification. We also show that student knowledge progression from the university data set can be linked with Udacity and Coursera datasets by giving an example.

Evaluation of RQ2

We have provided examples of correctly identified and incorrectly identified entities in the low-performance subgraph of the students in Student KG. The effectiveness of the Knowledge gap analysis was evaluated by identifying the percentage of students who scored low in the first-semester course and also scored less than 50% in the related course in the second semester. The relationship of the courses was determined using entity linking.

Evaluation of RQ3

We have evaluated the effectiveness of course recommendations by Student KG over Domain KG which involve layer 2 and layer 3 in our system. The students who took the same courses may face difficulty with various topics. So the priority of course recommendations should be different for each student. So, the personalization of the course recommendation could be identified by computing the diversity of course recommendations to the students. So, the diversity of the recommendations was computed as the dissimilarity (1-cosine similarity) between all pairs of recommendations. Normalized Discounted Cumulative Gain (NDCG) is a popular evaluation mechanism for evaluating recommendation systems (Kunaver & Požrl, 2017). Discounted cumulative gain (DCG) is a measure of ranking quality often presented in its normalized form known as Normalized DCG (nDCG or NDCG). NDCG is often used to measure the effectiveness of search engine algorithms and related applications. In our research, we recommend remedial courses that are related to the knowledge gaps. So, the relevance of recommendations was computed using NDCG. Gain is the relevance of each recommended item.

If there are ‘w’ weak topics of the student ‘S’ present in a MOOC course and there are ‘t’ topics from the courses the student ‘S’ has taken is present in the same MOOC course, then the relevance of the recommended item is computed as (w/t). If the relevance of the recommended item (I) for a student ‘S’ is G, the Cumulative Gain (CG) for K recommendations for all students is:

$$CG_{@K} = \Sigma_{i=1}^{K} G_i$$

Discounted Cumulative Gain (DCG) is:

$$DCG_{@K} = \Sigma_{i=1}^{K} G_i / \log_2(i+1)$$

NDCG is:

$$NDCG_{@K} = DCG/IDCG$$

$$IDCG_{@K} = \Sigma_{i=1}^{K_{Ideal}} G_{Ideal}/\log_2(i+1)$$

We have considered the relevance of the top 4 recommendations in our experiment. So the value of parameter K is 4. The denominator is the ideal DCG (IDCG) score when the most relevant items are recommended first.

Results

Figure 4 generated by GraphDB, shows a fragment of 3-layer student knowledge representation using SKR. The evidence for the high performance of a student in the KC 'Learn to express tasks recursively' is depicted in the figure. The mentioned KC node and two other KC nodes right to it are connected to the subject 'Recursion'. The nodes on the left side of the mentioned KC node are self-paced courses that also teach the topics related to 'Recursion'. The student node and performance nodes are connected using the 'hasEvidenceFor' relation where the Head node is the student and the Tail node is Performance. The graph shows that student performance data can be connected to the domain knowledge of multiple courses or learning systems without affecting their domain knowledge structure.

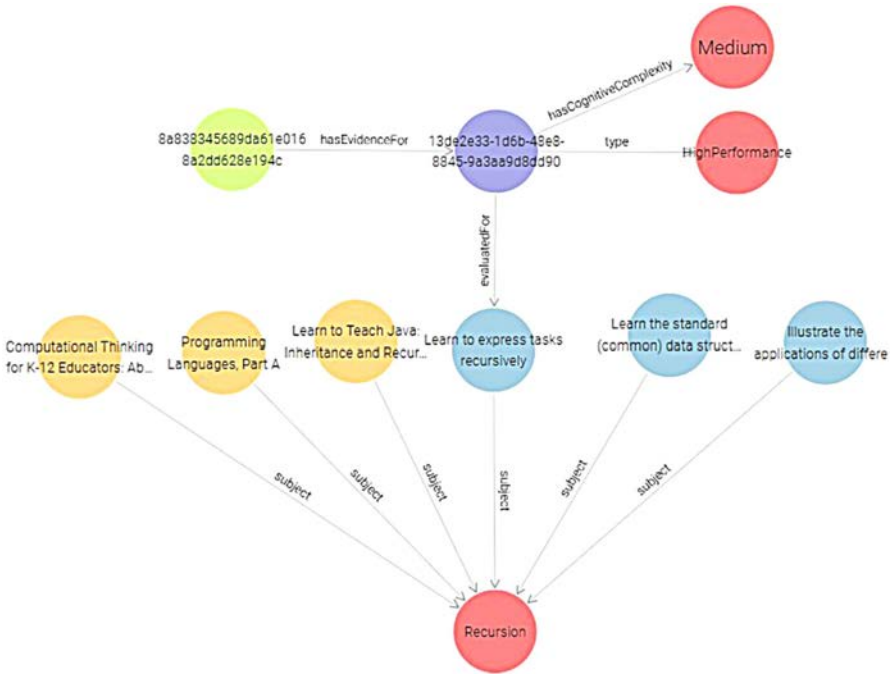


Fig. 4 Student knowledge evidence of a KC which has its subject 'Recursion', across the courses from various learning systems

Effectiveness of Entity Linking using CSO and DBpedia

The Course *Application of Machine Learning* contains 7 learning objectives with 44 KCs. The number of correctly and incorrectly identified entities is provided in Table 4. The precision and recall are computed from these values. The precision and recall of linking the *Application of Machine Learning* course to the knowledge layer is given in Table 5. The results show that CSO has better performance compared to DBpedia.

Figure 5 shows the connection between two Natural Language Processing courses using CSO as the mediator between them. Not all entities and relations are shown in the picture. The KCs of each course are shown on the left and right sides of the figure. Some of the topics from CSO which are linked with both the courses are shown in the middle of the figure.

Since the descriptions of KCs did not have enough details to generate the topics, the exam questions belonging to the KC were used as input to the entity disambiguation. The questions may contain mentions from different domains which ended up in producing entities that belong to various domains from DBpedia. Table 6 shows the examples of entities generated for KCs from CSO and DBpedia. Some of the entities are valid entities whereas some of them are not valid. CSO is generating topics closer to course or computer science since it contains computer science topics. DBpedia generates the exact topics along with the topics from other domains.

For example, in the 3rd row of Table 6, the CSO classifier generates the entities related to the course name. In contrast, the DBpedia spotlight generates entity equivalents that are not related to the computer science domain since the questions have other real-world entities.

Effectiveness of Identifying Knowledge Gaps Using Student KG

Table 7 has given examples of the topics in which some students are weak and the number of evidence for low performances of the students associated with those topics. Some wrong topics are detected by the CSO ontology listed in gray color in Table 7. The CSO ontology was created to organize the research knowledge. So it lacks basic concepts like *computational thinking*, *data structure*, or *algorithms*. CSO classifier classifies the text containing the concept *algorithm* into 'Adaptive algorithms'. The third row shows a different scenario. The CSO detects two major concepts and DBpedia detects another two major concepts. Even though the DBpedia spotlight detects natural language processing concepts, it could not link the parent topic. Since Data mining and Machine learning techniques are overlapping, and the student was enrolled in the

Table 4 The correctly and incorrectly identified topics from DBpedia and CSO

Expected correct entities	Correct entities identified (DBpedia)	Correct entities identified (CSO)	Incorrect identified entities (DBpedia)	Incorrect identified entities (CSO)
44	28	29	12	3

Table 5 The precision and recall values for the application of machine learning course

Precision (DBpedia)	Precision (CSO)	Recall (DBpedia)	Recall (CSO)	F1 Score (DBpedia)	F1 Score (CSO)
0.70	0.9062	0.6364	0.659	0.667	0.763

course 'Machine learning and Data mining', both of the concepts can be considered as correct. We analyzed whether the identified knowledge gaps have any effect on the future results of the students in the courses that are related to the knowledge gaps. 73.5% of students scored less than 50% for the courses that were related to their knowledge gaps in the previous semester.

We extracted the weakest topics of the students by identifying the topics with the maximum number of low performances and compared their results in related courses. Among the 200 out of 1381 students who had a topic with at least three low performances were selected for the comparison. 73.5 % of them scored less than 50% in the new courses with related topics.

Effectiveness of Student KG in Remedial Course Selection

The self-paced courses (course name, description, and KCs only) from Udemy and Coursera datasets were uploaded in Moodle LMS. We created a new field called Knowledge Profile ID in Moodle. The unique "Student ID" from the student KG is designated as the "Knowledge Profile ID" of the student when the student registers on Moodle. As the student logs in, a list of courses, aggregated by Algorithm 5, is displayed as the recommended list. Our approach equips Moodle to utilize the Student KG created by the university education system to make recommendations in its own system, thus enabling the sharing of student knowledge progression across heterogeneous learning systems.

We have compared the recommendations generated by the Domain KG and Student KG where the Student KG has the performance states of the student related to the

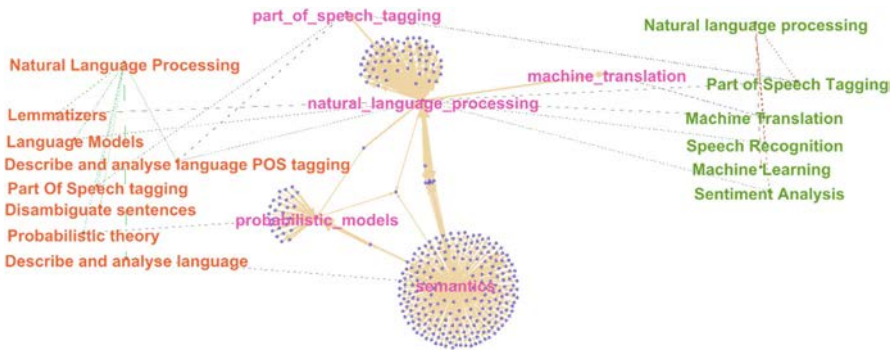


Fig. 5 Connection between two natural language processing courses from two learning systems using CSO ontology classification (CSO ontology topics are shown in the middle. All connections are not shown)

Table 6 The samples of KCs from various courses and their mapped entities generated by the CSO classifier and DBpedia spotlight

Course	KC	CSO	DBpedia
Machine learning and data mining	Linear regression	Correlation analysis, Regression model, big data, data analytics, machine learning	Regression analysis, machine learning, logistic regression, curve fitting, statistical hypothesis testing, loss function, linear regression
Applications of machine learning	Clustering and its applications. Learn various types of clustering	K-means clustering algorithm, dendrogram, clustering methods, hierarchical clustering, clustering algorithms, machine learning	BFR algorithm, machine learning, hierarchical clustering, k-means clustering, time complexity, starship development history, cluster analysis, clustering high-dimensional data, distance matrix, mahalanobis distance
Natural language processing	Gain knowledge about the basic probability theory	Markov model, probability, natural language processing, transition probabilities, hidden Markov model (hmm), hidden Markov models	Entropy, Markov model, Algorithm, ice, heart sounds, logical conjunction, probability theory, probability, hidden Markov model

domain knowledge. We considered the first four recommendations for all students for computing the performance metrics. We computed the diversity and relevance of the recommendations which shows the personalization of recommendations. The diversity and relevance scores have been computed for the students enrolled in the same batch and for all students. The results are given in Table 8. All students who enrolled in the same batch and attended most of the same courses were given similar

Table 7 The weak topics of students and the count of low-performance evidence for the same

Student	Topic from CSO (Count of low performances)	Topic from DBpedia (Count of low performances)
s1	Combinatorial problems (5)	Computational thinking (5)
s2	Object-oriented Programming (7) Object-oriented design (7) Adaptive algorithms (5)	Object-oriented Programming (7) Data structure (7) Algorithm (5)
s3	Natural language processing (7) Data Mining (6)	Machine learning (7) Data structure (5)

Table 8 The personalization scores for the recommendations generated by the student KG and domain KG

Performance metrics	Students in the same batch (most courses are same)	All students who various enrolled in various courses
Diversity-student KG (%)	16.9	88.28
Diversity-domain KG (%)	18.6	79.24
NDCG-student KG (%)	98.9	81.45
NDCG-domain KG (%)	69.8	76.96

recommendations, so there is less diversity. The Cumulative Gain is computed based on the weak topics of the students. So, a higher NDCG means higher relevance of the recommendation to the knowledge gap of the students and vice versa (See section 3.5.3). So, the results show that Domain KG provides the first four recommendations that are less relevant to the knowledge gaps of students.

Figure 6 shows the visual explanation generated from the low-performance subgraph which justifies the remedial recommendations. It shows the topics in which

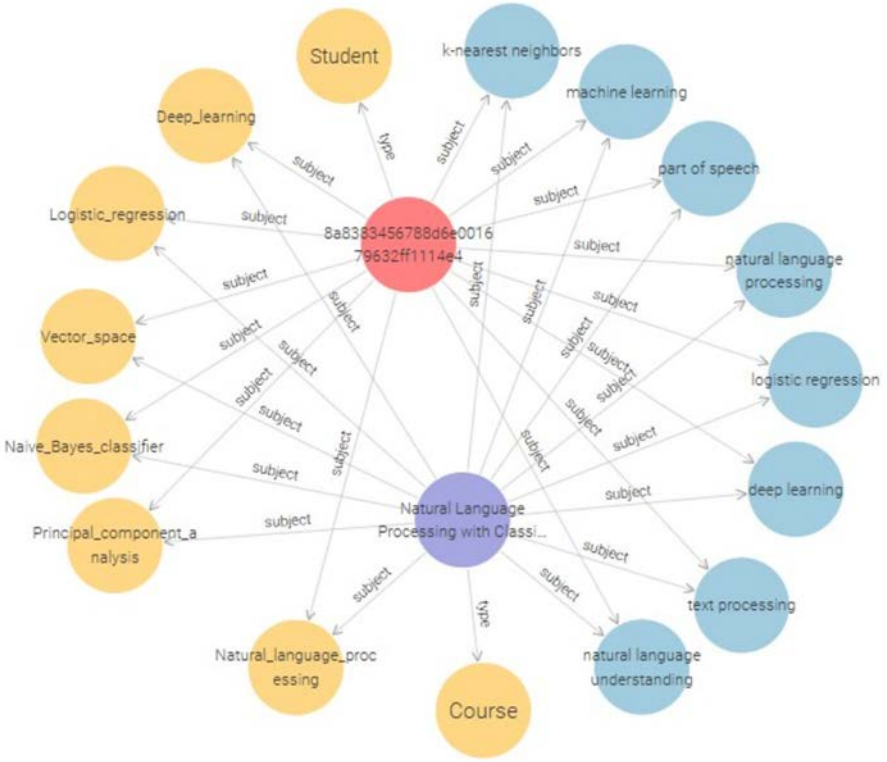


Fig. 6 Explanation generated for remedial course recommendation from the low-performance subgraph of a student

the student has evidence for low performance and the remedial course's relation with those topics. The node on the right side of the diagram shows the Student node and the node on the left side is the self-paced Course. The nodes that have the *subject* relation with the Student node and Course node explain the topic recommendation. The student has low CCSS scores in the subjects and therefore SKG recommends the self-paced course that is related to these topics.

Discussion

In this section, we discuss the effectiveness and research opportunities of the 3-layer knowledge representation model for guiding students across heterogeneous learning systems. We tried to mitigate the difficulty by not revealing proprietary details, as learning systems seldom interoperate due to differences in methodology, standards, and privacy policy. For relating the KCs across learning systems in our model, the KCs are connected to the knowledge layer which avoids the necessity of direct communication between learning systems. The necessary data can be maintained as a knowledge graph which could be queried by the web. The key part in relating the domain knowledge of the learning systems is entity linking with the knowledge layer.

RQ1: Effectiveness of Entity Linking in Relating the Domain Knowledge of Heterogeneous Learning Systems Using the Knowledge Base

In this article, we discussed entity linking using both domain ontologies and knowledge bases. If the learning systems directly connect the domain knowledge of every other system, then a KC classification or annotation mechanism is required in every learning system. In the case of direct entity linking, each KC in each learning system will have links to all related topics in the other two learning systems. Using the knowledge base as the mediator is effective because it avoids the necessity of having a classification mechanism in every learning system. It reduces the number of entity links per KC.

In knowledge layer linking, the precision, recall, and F1 score measures tell that CSO has better performance for the machine learning course. The domain ontologies describe the relationship between educational topics which helps to analyze the main area of studies in which students lack skills. CSO was the computer science domain ontology utilized for entity linking which was trained on academic research papers. So, basic educational concepts like data structures and algorithms were missing in the domain ontology. However, there was no better choice since our research was focused on computer science courses. DBpedia contains most of the topics from the computer science domain, however, the relationships of the linked resources were not limited to educational concepts. The subject of resources can provide the relevant concepts along with the irrelevant ones according to the educational context. Context awareness needs to be implemented while using entity linking with DBpedia. So identifying the relationships between appropriate concepts requires more computational effort. Creating an ontology or knowledge base of all educational concepts could be researched further. ConceptNet has been created to represent the concepts from various domains

(Speer et al., 2017). Since it is designed to represent general knowledge, the advanced topics from computer science were not present in ConceptNet. The input text for the entity disambiguation process is also important as the mentions of topics or objects out of the learning context could be chosen for entity linking especially while using the knowledge bases (Table 6, row 3). Correct and precise text as input for entity linking provides better results.

RQ2: Effectiveness of Student KG in Analyzing the Knowledge Gap of Students

Our analysis has shown that 73.5% of students failed later in other courses related to the topics in which the student had at least 3 low performances earlier. The remaining 26.5% of students succeeded in the courses related to the knowledge gaps. The effect of the knowledge gap was investigated in the immediate future as the time frame between the exams was six months. So we assume that the remaining students could fill their knowledge gaps during the learning period of the new courses. Another reason could be because the influence of the weak topics of the students on the new courses was low. Since the CSO classifier successfully linked the KCs to the parent concepts, it helped in identifying major areas of education where the students were weak. DBpedia contributed to identifying the lack of knowledge in a particular topic.

RQ3: Effectiveness of Student KG in Remedial Course Recommendation

We have shown that remedial courses can be recommended by querying the Student KG. Students receive better personalization on the ranking of remedial recommendations when the knowledge states of students are included in the knowledge graph. The Domain KG fails to generate the relevant recommendations with the highest rank since it does not know the requirements of the students. The Domain KG can generate more useful recommendations if the student is looking for a remedial recommendation for one course or one topic. The recommendations will have better relevance in that case even if the diversity of the recommendations will be low. The knowledge analysis and remedial recommendations for a student are solely dependent on the learning history of that student and do not require the details of other students. So, the number of students is not a concern in our model for making relevant recommendations, unlike the training-based models. The recommendations can also include new courses that do not have any usage history since our model is dependent on the interrelation between KCs.

Limitations and Opportunities

Our research work has shown that linking the domain knowledge to the knowledge bases will help relate the courses from various educational systems. In our research, we employed DBpedia and CSO as the knowledge bases. CSO has performed better than

DBpedia. However, all topics in the selected courses were not present either in DBpedia or CSO. This points out the need to have the knowledge base of educational concepts which may produce better results in entity linking. Another limitation of our approach is not having the fine-grained KCs and input text to the classification APIs that describe each KC with the concepts employed in the corresponding lessons. The text in the questions was not good enough in some courses to mention the concepts corresponding to the learning objectives. DBpedia and CSO have their topic classification algorithm which sometimes fails to detect the correct educational concept. For example, DBpedia identified Python as a snake instead of identifying it as the programming language. Two courses with Python in its syllabus were linked to the same topic, which is not an educational concept. The linking was incorrect even though the courses were linked with each other. So, to bring this approach into practice, a better dataset, educational knowledge base, and a classification algorithm that can detect educational concepts are required.

Conclusion

In this paper, we have discussed the 3-layer knowledge graph for representing the knowledge progression of the students across various learning systems. The performance evidence for KCs acts as the top layer of our framework, the KCs across the learning systems act as the middle layer and the third layer consists of the links between KCs. We designed SKR ontology to define the relations in the proposed knowledge graph. We have discussed the possibility and benefits of connecting heterogeneous learning systems by employing domain ontologies and knowledge bases as the mediators between learning systems. We show that using such a mediator for connecting the learning systems reduces the required amount of linking operations.

The knowledge progression of university students was represented using the 3-layer knowledge graph. The domain knowledge of computer science courses at a university and the self-paced courses from two MOOC systems were also represented in the same knowledge graph using SKR ontology. We utilized the CSO classifier and DBpedia spotlight for entity linking/entity disambiguation to identify the educational concepts corresponding to the domain knowledge of various learning systems. We found both techniques generated relevant topics when the KC descriptions were precise and clear. The CSO classifier identifies the parent concepts along with the educational topics whereas the DBpedia spotlight focuses on identifying the exact topics from the text. CSO contains advanced research topics and lacks basic educational concepts which prevents it from generating suitable entities for basic computer science courses. In our experiment, CSO performed better than DBpedia because CSO has educational topics from and only from Computer science.

We discussed the knowledge gap analysis and remedial course recommendations to demonstrate the applicability of our model. We verified the correctness of the knowledge gap analysis by comparing the results of the eventual courses related to the weak topics of the students.

We compared the remedial recommendations generated by Student KG which contains the performance evidence along with domain knowledge and the Domain KG which contains the domain knowledge only. The remedial recommendation generated by Student KG had having better order of ranking than the recommendations produced by Domain KG. Graph embedding models are popular in research related to KG. Since student KG is dynamic, we recommend further research on inductive learning models for predictive tasks using Student KG. Since KGs are popular instruments in causal models, future research could investigate the benefits of Student KG in various reasoning tasks.

Author Contributions Both of the authors contributed to the study's conception and design. The data collection and implementation were done by Soumya MD. The first draft of the manuscript was written by Soumya MD and ShivSubramani Krishnamoorthy commented on previous versions of the manuscript. Both authors read and approved the final manuscript.

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Data Availability The university dataset is private dataset which is not available. The links to the MOOC datasets are provided as references.

Declarations

Conflict of Interest The authors have no relevant financial or non-financial interests to disclose.

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